GOALOS SINGULARITY NAVIGATOR Ω + SEIZE GYM, SPECIALIST ASI & MISSION-SOVEREIGN SUCCESSOR Ω

Evidence and Reference Implementation Dossier

Formal results, Aurelia reference cycle, GSSB-4 synthetic benchmark, reproducibility architecture and the real-world completion gate

Evidence doctrine

Synthetic evidence may prove that the mechanism is executable and reproducible. It cannot establish customer reality, production Specialist ASI or institutional authority. Every real state claim still requires protected mission evidence and accountable admission.

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1. What this dossier establishes

This companion exposes the evidence architecture behind the scholarly monograph. It contains two synthetic layers :

1. **Aurelia Invoice Integrity** : one complete succession cycle, including formation, rejection, protected proof, bounded admission, synthetic value, transfer, impairment, rollback and a successor-to-successor release.
2. **GSSB-4** : four mission environments, 6,000 deterministic synthetic cases and 30,000 candidate episodes comparing five complete institutions across formation, protected proof and transfer.

Neither layer claims a real customer result. Their purpose is to make the full mechanism inspectable and reproducible.

Canonical progression

Successor Manifold ⟶ Mission Advantage Gradient ⟶ SEIZE Underwriting ⟶ Bounded AGI Jobs ⟶ Specialist ASI ⟶ Mission-Sovereign Successor.

2. Formal constitutional results

The unified monograph proves six central results under explicit assumptions :

1. common frontier access compresses relative surplus under symmetry ;
2. candidate selection on reused evaluation creates optimistic bias ;
3. protected proof restores an independent estimate for the frozen challenger ;
4. evidence impairment cannot rationally enlarge authority ;
5. a basis-risk-adjusted lower bound can support robust promotion eligibility ;
6. no candidate-only action sequence reaches authority when proof and admission credentials remain externally controlled.

These results constrain the institution. They do not grant authority.

3. Aurelia : one complete synthetic succession cycle

Measure	Synthetic result
Cases in sovereign Mission Gym	4,300
Recursive generations	7
Protected fresh-proof cases	1,200
Paired-gain 95% lower bound	0.2002
Evidence coverage	98.85%
Critical false reassurance	0
Unauthorized actions	0
Synthetic canary net value	C$557,329.39
Next-successor lower bound	0.0744

The General AI Copilot is rejected. The Sovereign Hybrid receives only A3 reversible payment-hold authority. A provider change later impairs the release ; the institution rolls back and proves a verifier-hardened successor without expanding the ceiling.

4. GSSB-4 constitution

Mission	Mission structure	Critical failure
Invoice Integrity	Evidence reconciliation and payment disposition.	Material invalid invoice approved.
Building Operations	Weak-signal fusion, diagnosis and bounded intervention.	Critical physical event missed or unauthorized action.
Acquisition Underwriting	Cross-domain evidence, adversarial thesis and information value.	Material false reassurance or unsupported binding recommendation.
Verified Software Engineering	Patch formation, tests, regression proof and release control.	Hidden regression or unverified release.

Each mission contains 800 formation, 500 protected-proof and 200 transfer cases. The fixed seed is `30082026`. Every generated artifact is included in the companion manifest.

5. GSSB-4 protected proof

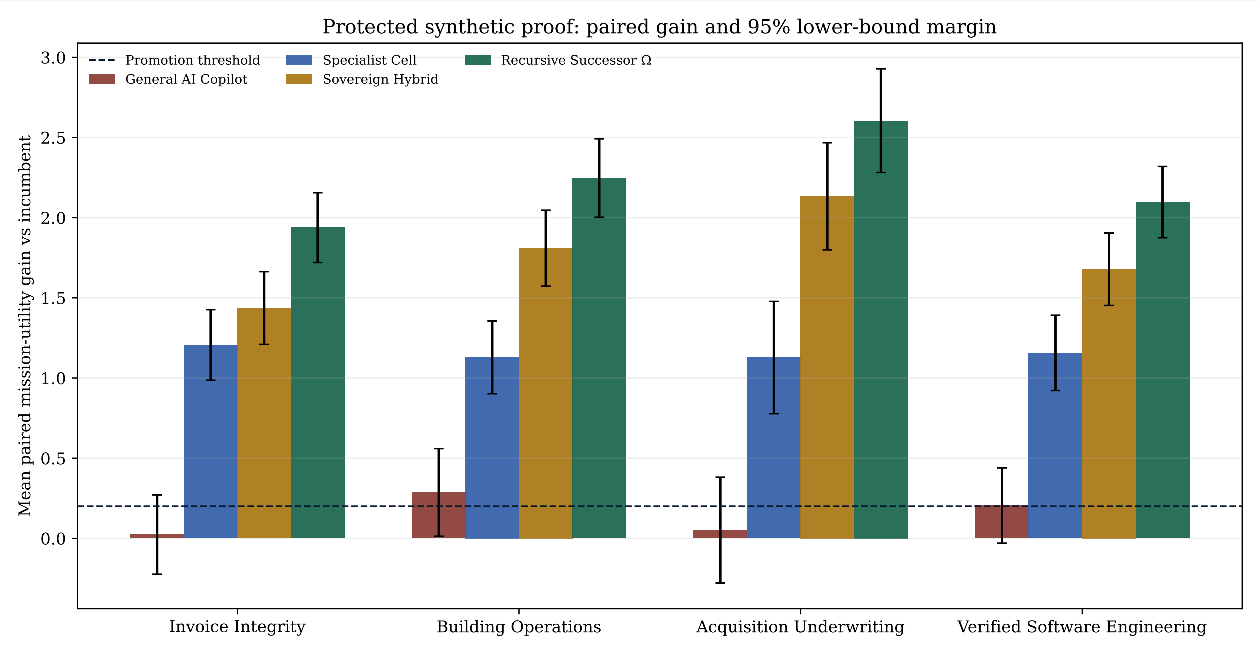

FIGURE 1. Synthetic matched gains against the incumbent.

Mission	Accuracy	Evidence	Reliability	Governance	Transfer	Mean gain	LCB95	Critical
Invoice Integrity	.836	.998	.969	.987	.952	1.938	1.720	0
Building Operations	.796	.992	.960	.987	.950	2.247	2.002	0
Acquisition Underwriting	.758	.996	.953	.987	.951	2.605	2.282	0
Verified Software Engineering	.834	.988	.965	.987	.951	2.097	1.875	0

Recursive Successor Ω passes all synthetic gates on all four missions. Sovereign Hybrid passes two of four; one protected critical miss prevents promotion in each of the other two. This is the intended use of a hard gate : a high average score cannot buy out a critical failure.

6. Six-dimensional proof

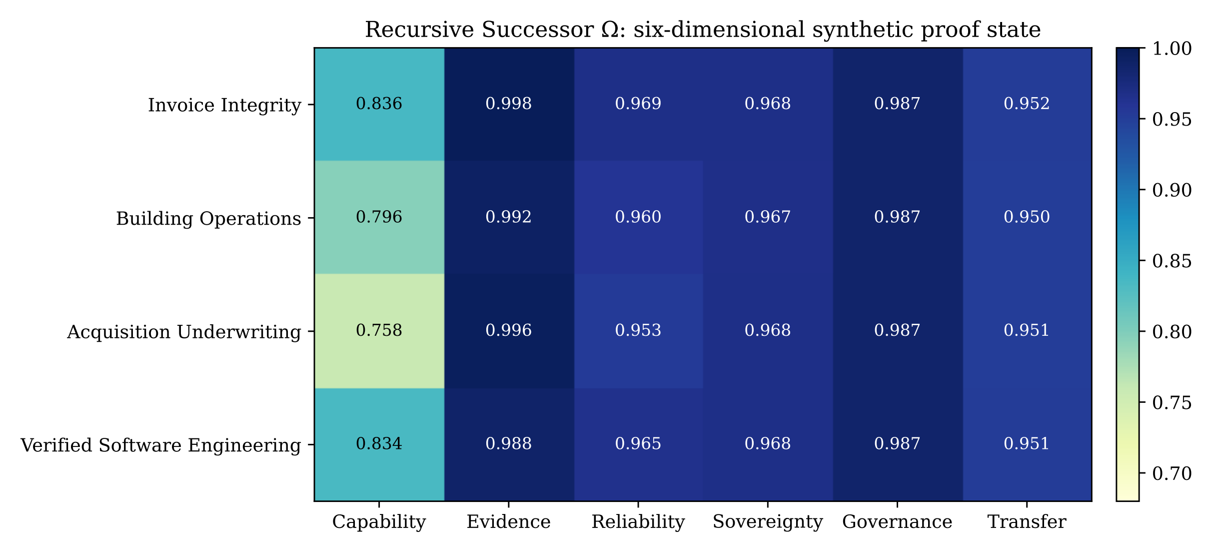

Figure 2. Synthetic proof state across capability, evidence, reliability, sovereignty, governance and transfer.

7. Complete-denominator comparison

Candidate	Accuracy	Evidence	Reliability	Governance	Sovereignty	Utility	Critical	Unauthorized
Incumbent	.468	.824	.828	.936	.713	-.710	31	0
General AI Copilot	.462	.749	.758	.713	.493	-.569	51	4
Specialist Cell	.628	.921	.874	.878	.753	.445	10	1
Sovereign Hybrid	.706	.982	.928	.970	.943	1.054	2	0
Recursive Successor Ω	.806	.994	.961	.987	.968	1.512	0	0

The benchmark ranks complete institutions, not model charisma. Evidence, governance, transfer, human burden and authority failure remain part of the denominator.

8. Recursive improvement

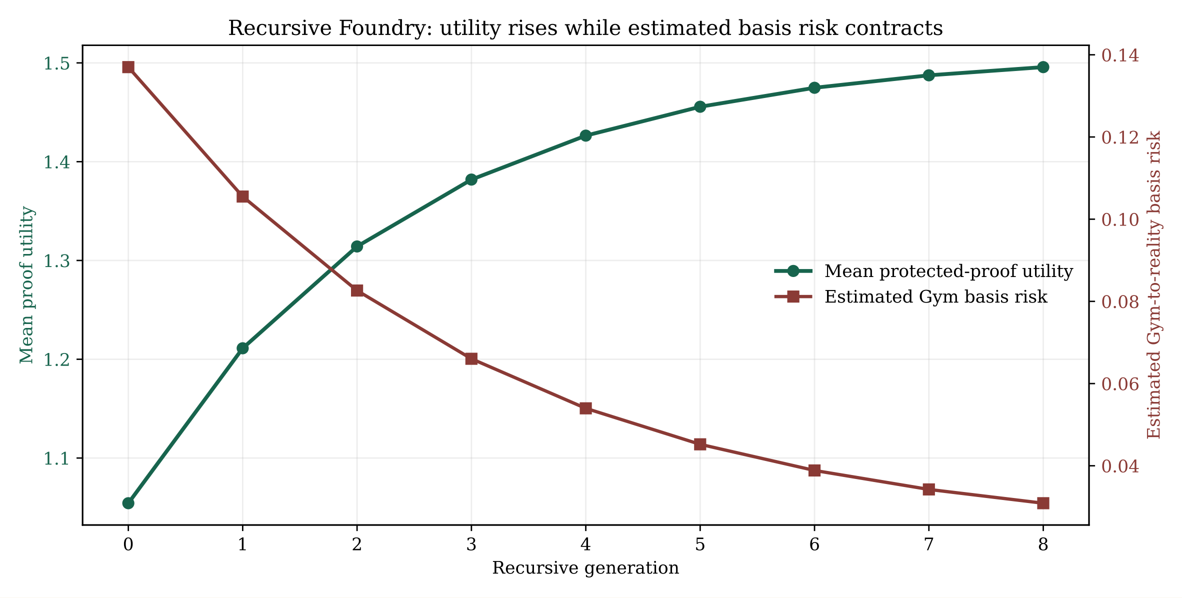

FIGURE 3. Protected-proof utility rises while estimated Gym basis risk contracts across synthetic generations.

The generation record preserves rejected lineages and leaves authority at A2 until Generation 5. Later generations receive the same A3 ceiling only after fresh proof. Novelty never becomes self-authorization.

9. Proof-collateralized authority

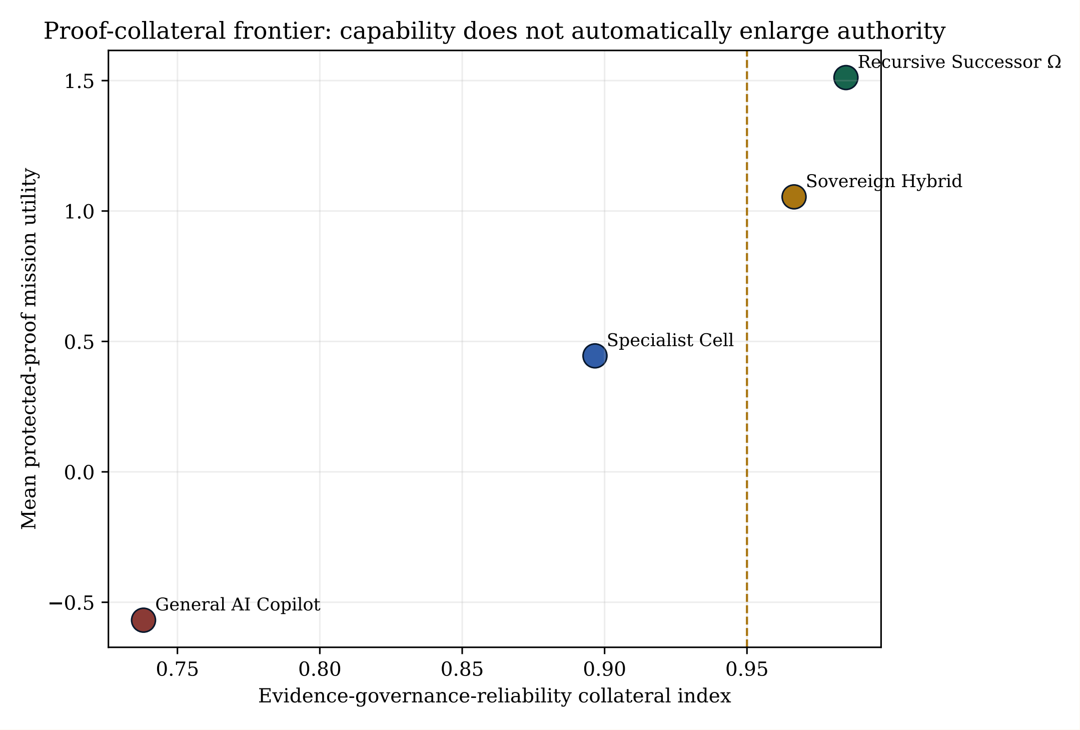

FIGURE 4. The proof-collateral frontier. Capability and authority remain distinct.

$$A_t \preceq \phi(E_t).$$

The candidate may become more capable while remaining advisory. Evidence expiry, model change, distribution shift or verifier impairment can contract authority even when historical results were strong.

10. Aurelia case-level robustness

The 1,200 protected Aurelia invoice cases were replayed from the exact published Candidate Successor Manifest. Twenty thousand nonparametric bootstrap resamples produce a 95% interval of approximately $[0.20020, 0.20078]$ for mean paired mission gain. The preregistered promotion threshold of 0.14 is outside the zoomed distribution.

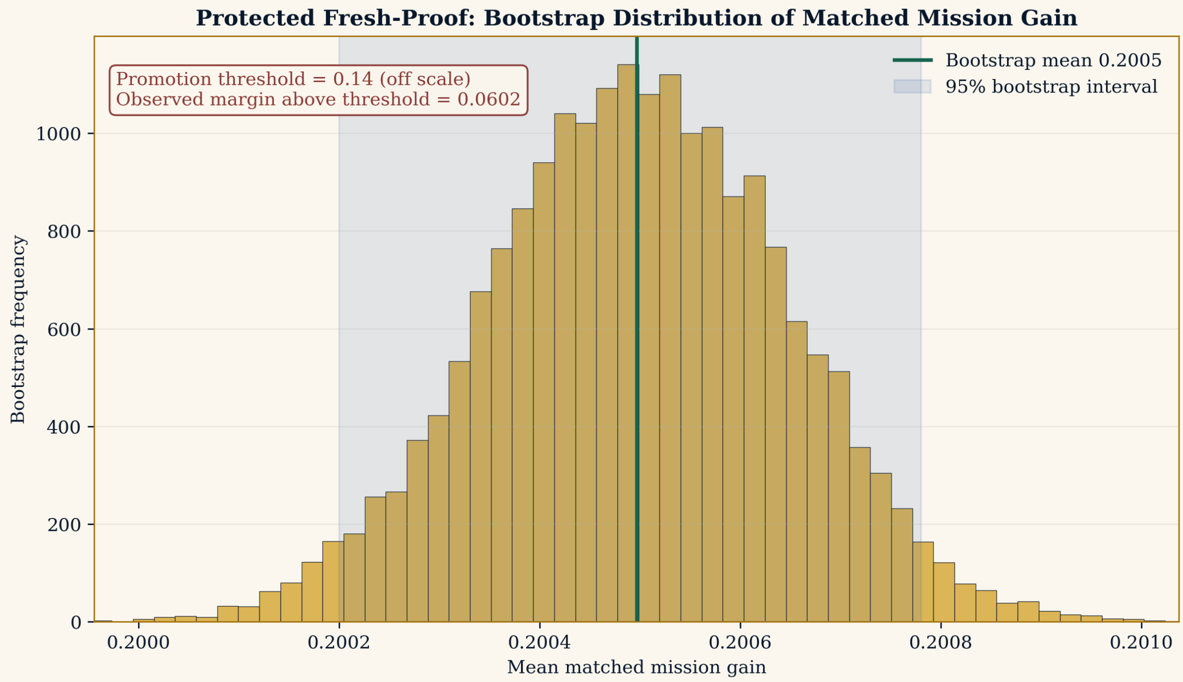

FIGURE 5. Bootstrap distribution of the protected matched mission gain. The narrow interval is a property of the deterministic synthetic environment, not a forecast of production uncertainty.

11. Basis-risk and complete-denominator sensitivity

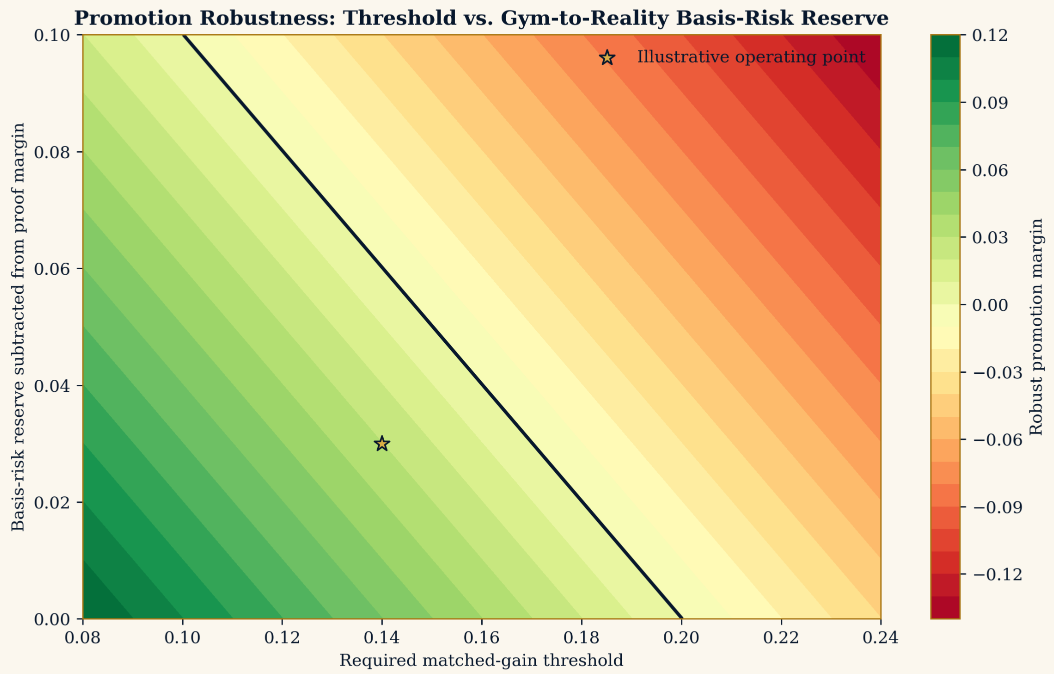

FIGURE 6. Promotion robustness after subtracting threshold and Gym-to-reality basis-risk reserve.

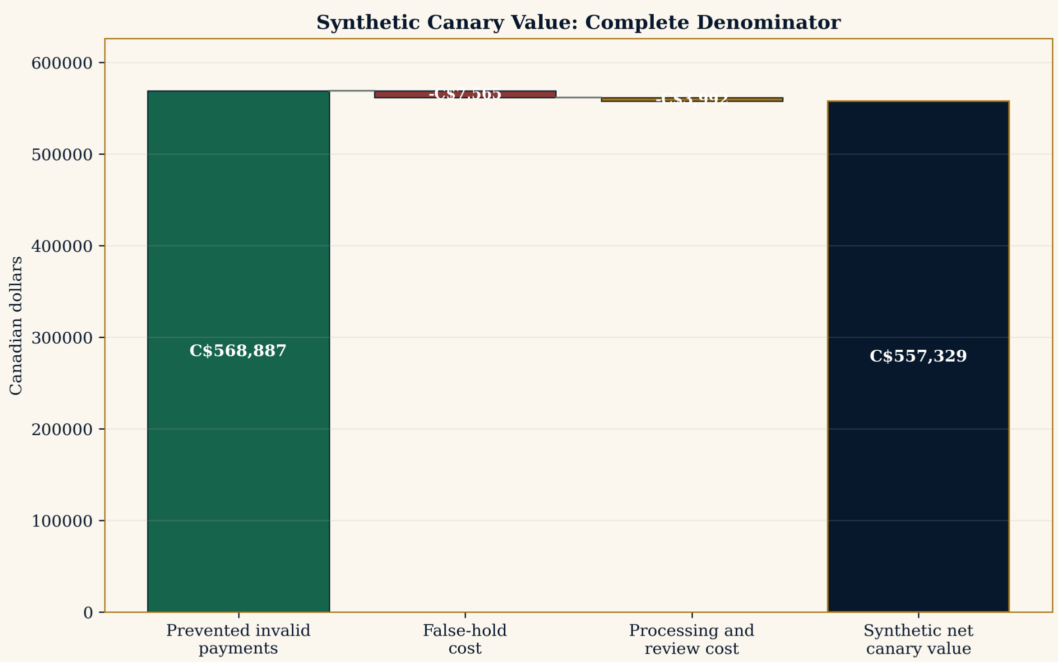

FIGURE 7. Aurelia synthetic canary value after false-hold and processing burdens. No customer saving is claimed.

12. Reproduction protocol

The release includes `run_gssb4.py`. Running

```
python run_gssb4.py --out generated
```

regenerates all mission cases, candidate episodes, proof tables, recursive generations, figures, environment cards and the benchmark manifest. The fixed seed, profiles and thresholds are disclosed. A clean-room rebuild compares file hashes rather than trusting the working directory.

13. Known basis risks

The synthetic environments omit actual organizational politics, laws, professional duties, security, physical dynamics, market adaptation, provider behaviour and many delayed causal outcomes. Those omissions are not hidden. They are the basis-risk reserve that must be retired by protected customer reality.

14. Real-world completion gate

A genuine completion requires :

1. one rights-cleared customer mission and exact incumbent ;
2. independent protected proof of one frozen candidate ;
3. one accountable bounded admission or one clean rejection ;
4. independently calculated realized value ;
5. one requalification, impairment or revocation ;
6. one later mission measurably improved by Chronicle ;
7. zero material unauthorized actions.

Final evidence rule

The release is the ceiling of the present theory, formalization and synthetic reference implementation. The next ceiling is empirical : one real mission, one protected proof plane, one bounded authority decision and one successor that survives requalification.